

Understanding Artificial Intelligence and Machine Learning

A simple learning guide

AI and machine learning get mentioned everywhere, and for a while I used the words as if they meant the same thing. They don't. While studying this topic I realised that most of the confusion goes away once you see how the ideas fit inside each other. This guide walks through the main terms in the order that made the most sense to me, with one everyday example for each.

1. Artificial Intelligence (AI)

Artificial intelligence is about making machines do things that normally need human thinking, such as understanding speech, recognising faces, making decisions or translating a language. The machine isn't "thinking" the way we do. It just behaves in a way that looks intelligent for a specific job.

AI is the big umbrella term. Some AI systems follow rules that a programmer wrote by hand, and others learn from data. That difference is what leads to machine learning.

Example: A voice assistant like Google Assistant or Siri. You ask it to set an alarm, it works out what you said and does it, which is something only a person could do in the past.

2. Machine Learning (ML)

In normal programming, I write exact instructions for the computer. With machine learning, I give the computer a lot of examples and let it work out the instructions by itself. It looks through the data, finds patterns, and then uses those patterns to make predictions or decisions on new data it hasn't seen before.

ML is a part of AI. All machine learning counts as AI, but not all AI uses machine learning. A chess program that only follows hand-written rules is AI without being ML.

Example: The spam filter in your email. It has seen thousands of emails already marked as spam or not spam, so it picks up on things like suspicious links or words such as "free prize" and uses them to judge new emails.

3. Deep Learning (DL)

Deep learning is a type of machine learning that uses neural networks, which are loosely inspired by the way brain cells connect. A network is made of layers of small units. Each layer takes in information, does a little processing and passes it on to the next one. "Deep" simply means there are many layers.

Early layers pick up simple things and later layers combine them into bigger ideas. In a photo, the first layers may notice edges, the next ones shapes, and the last ones put it together as "this is a cat". Deep learning needs a lot of data and computing power, but it does very well on messy things like images, sound and language.

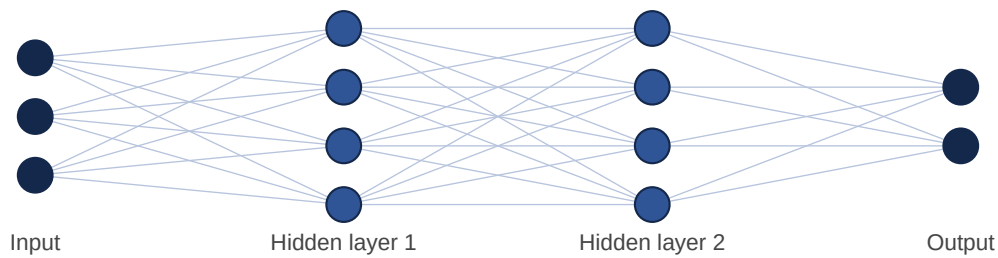


Figure 1: A simple neural network. Information flows from left to right through the hidden layers.

Example: Face unlock on a phone. A deep network learns the features of your face well enough to tell you apart from other people, even with glasses or in different light.

4. Supervised Learning

Supervised learning is like studying with an answer key. The computer is given data where the correct answer is already attached, called labelled data. It makes a guess, checks it against the right answer, and adjusts itself until its guesses get close. After enough practice it can handle new examples on its own.

It is mostly used for two kinds of problems: classification (choosing a category, such as spam or not spam) and regression (predicting a number, such as a price).

Example: Predicting house prices. The model is trained on past sales that include the size, location, number of rooms and the actual selling price. Once it has learned how those details relate to the price, it can estimate the price of a house that hasn't been sold yet.

5. Unsupervised Learning

Here there is no answer key at all. The computer only gets raw data and has to find some structure in it by itself, usually by grouping similar things together (clustering). Nobody tells it what the groups should be, so it decides based on what is similar in the data.

This is useful when we don't know exactly what we are looking for. The results often need a human to look at them and decide what the groups actually mean.

Example: A shopping app grouping its customers. Based on what people buy and how often, the system might find clusters like "bargain hunters", "regular grocery buyers" and "gadget lovers", without anyone defining those groups beforehand.

6. Reinforcement Learning

Reinforcement learning works like training a dog or learning a video game. The learner, called an agent, tries an action in some environment and gets a reward if it was good or a penalty if it was bad. Nobody shows it the right move. It learns by trial and error, and over many attempts it figures out which actions lead to the most reward in the long run.

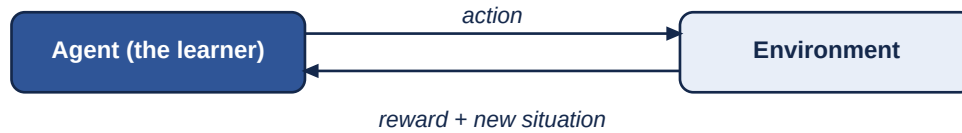


Figure 2: The reinforcement learning loop.

Example: A robot learning to walk. At first it falls over constantly. Each time it stays balanced it gets a reward, and each time it falls it gets a penalty, so it gradually settles on movements that keep it upright. The same idea is behind programs that learn to play games like chess and Go.

7. Difference Between AI, ML and DL

The easiest way I remember it is as circles inside circles. DL sits inside ML, and ML sits inside AI.

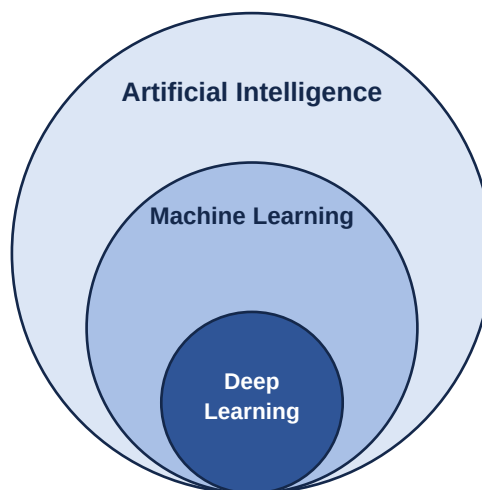


Figure 3: How AI, ML and DL are related.

Term	What it means	Relationship	Example
AI	Machines doing tasks that usually need human intelligence	The broadest idea	Voice assistant
ML	Machines learning patterns from data instead of fixed rules	A part of AI	Email spam filter
DL	ML that uses neural networks with many layers	A part of ML	Face unlock on a phone

8. Comparison of Learning Types

Type	Data used	How it learns	Example
Supervised	Labelled data (answers included)	Compares its guesses with the correct answers and corrects its mistakes	Predicting house prices
Unsupervised	Unlabelled data (no answers)	Looks for similarities and groups in the data on its own	Grouping shop customers
Reinforcement	No fixed dataset; feedback from actions	Tries actions and learns from rewards and penalties	Robot learning to walk

9. Conclusion

To sum it up, AI is the overall goal of making machines act intelligently, machine learning is one way of getting there by learning from data, and deep learning is a powerful kind of machine learning built on layered neural networks. Supervised, unsupervised and reinforcement learning are the three main ways a machine can learn, depending on whether it has answers, no answers, or only rewards.

What stood out to me while writing this is how often we already use these ideas without noticing, in spam filters, map apps, video recommendations and phone cameras. They are not perfect and they are only as good as the data they learn from, but they save a lot of time and make everyday tools more useful. Understanding the basics also makes it easier to judge what AI can realistically do and what it can't.